

DATA CLEANING AND PREPARATION

Project Report

A Python-based data cleaning, validation,
preparation and analysis project

Technology Stack

Python • Pandas • NumPy • Matplotlib

GitHub Repository

<https://github.com/raahulraj2007k-pixel/Data-Cleaning-and-Preparation>

1. Introduction

Data cleaning and preparation is an important stage of the data analysis process. Real-world datasets often contain missing values, invalid records, inconsistent formatting, duplicate records and incorrect data types. If these problems are not addressed, they can produce unreliable analysis results.

This project demonstrates a complete data cleaning and preparation workflow using Python. A synthetic customer sales dataset was created with realistic data-quality problems. The dataset was then cleaned, validated and prepared for analysis.

2. Project Objectives

- Load and inspect a raw CSV dataset.
- Identify missing, invalid and inconsistent data.
- Standardize text fields and column names.
- Validate email addresses.
- Convert multiple date formats into a consistent format.
- Handle invalid negative quantity and price values.
- Detect and remove business-key duplicate records.
- Fill missing values using appropriate strategies.
- Create a calculated total_amount field.
- Validate the cleaned dataset.
- Perform basic sales analysis and generate visualizations.

3. Dataset Description

The project uses a synthetic customer sales dataset created specifically for demonstrating data cleaning techniques. It does not contain private customer information or confidential records.

Property	Value
Original rows	30
Original columns	8
Final rows	29
Final columns	9
Input format	CSV
Output format	CSV

Dataset Columns

Column	Description
order_id	Unique order identifier
customer_name	Customer name
email	Customer email address
city	Customer city
order_date	Date of order
product	Purchased product
quantity	Quantity purchased
unit_price	Price per unit
total_amount	Calculated total sales amount

4. Identified Data Quality Issues

Issue	Count / Scope	Action Taken
Missing email	2	Handled using a placeholder
Missing order date	1	Filled using median date
Missing quantity	1	Filled using median
Missing unit price	1	Filled using median
Invalid email format	1	Converted to missing and handled
Negative quantity	1	Converted to missing
Negative unit price	1	Converted to missing
Inconsistent text formatting	Multiple	Standardized
Multiple date formats	Multiple	Converted to standard date
Business-key duplicate	1	Removed

5. Data Cleaning Methodology

1. Load the raw CSV dataset using Pandas.
2. Inspect dataset shape, columns, data types and missing values.
3. Standardize column names.
4. Remove leading and trailing spaces from text values.
5. Standardize city and product capitalization.
6. Convert email addresses to lowercase.
7. Validate email addresses using a regular expression.
8. Parse multiple date formats using Pandas.
9. Convert quantity and unit price into numeric data types.
10. Convert negative quantities and prices into missing values.
11. Detect business-key duplicates.
12. Remove duplicate business records while preserving the first occurrence.
13. Fill missing quantity and price values using the median.
14. Fill missing emails using a safe placeholder.
15. Fill missing order dates using the median date.
16. Create the total_amount calculated field.
17. Perform final validation and save the cleaned dataset.

6. Implementation

The implementation is divided into two Python scripts. The first script performs cleaning and preparation, while the second script performs analysis and visualization.

6.1 Data Cleaning Script

File: src/data_cleaning.py

- Loads data/raw_dataset.csv.
- Performs validation and transformation.
- Generates cleaning_log.txt.
- Creates data/cleaned_dataset.csv.
- Reports final missing values and duplicate counts.

6.2 Data Analysis Script

File: src/data_analysis.py

- Compares raw and cleaned datasets.
- Compares missing values before and after cleaning.
- Calculates descriptive statistics.
- Calculates product-wise sales.
- Calculates city-wise sales.
- Generates charts using Matplotlib.
- Creates analysis_summary.txt.

7. Cleaning Results

Metric	Before Cleaning	After Cleaning
Rows	30	29
Columns	8	9
Missing values	Present	0
Exact duplicate rows	0	0
Business-key duplicates	1	0
Invalid emails	1	0
Invalid negative values	2	0

The final dataset contains 29 rows and 9 columns. All missing values were handled, invalid values were corrected, and duplicate business records were removed.

8. Data Analysis and Visualization

After cleaning, the prepared dataset was analyzed to demonstrate that the resulting data can be used reliably for basic business analysis.

8.1 Product-wise Sales

Product	Total Sales
Laptop	439,000
Tablet	256,500
Mobile Phone	211,500
Headphones	20,800

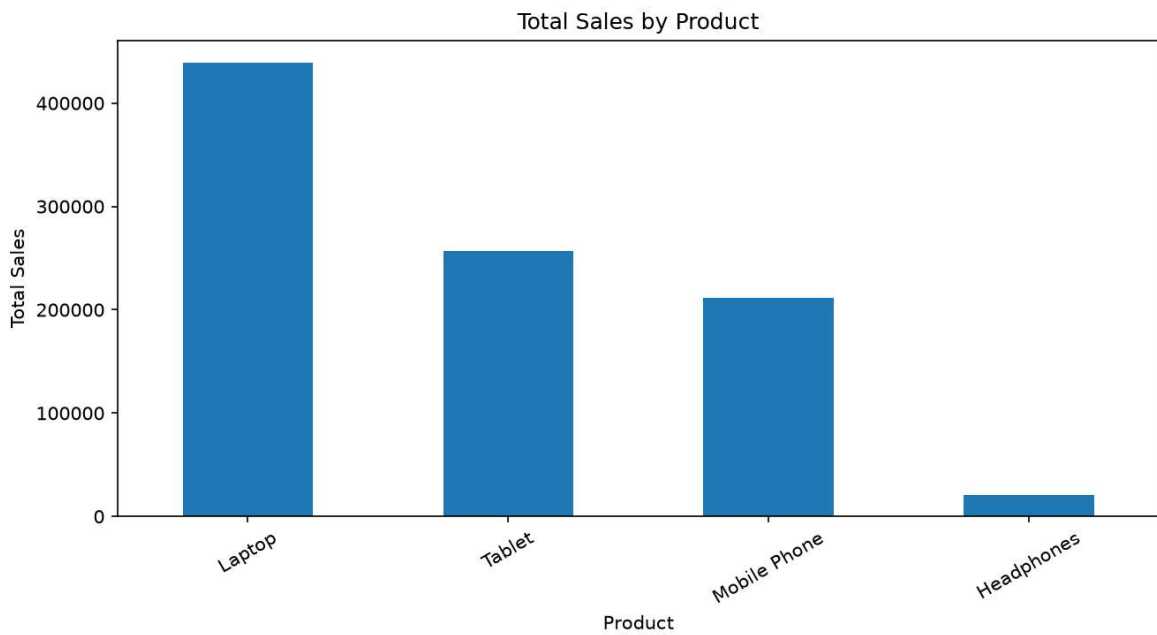


Figure 1: Product-wise sales visualization

8.2 City-wise Sales

City	Total Sales
Delhi	397,800
Bangalore	229,000
Mumbai	195,100
Kolkata	105,900

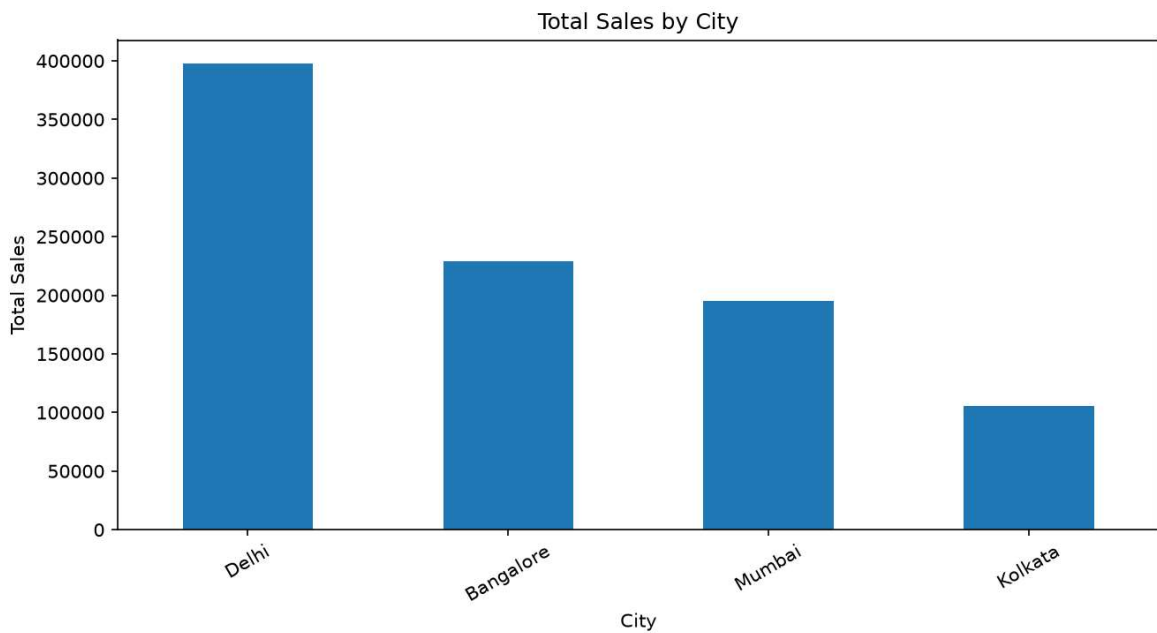


Figure 2: City-wise sales visualization

9. Data Quality Comparison

The following visualization compares missing values in the raw and cleaned datasets.

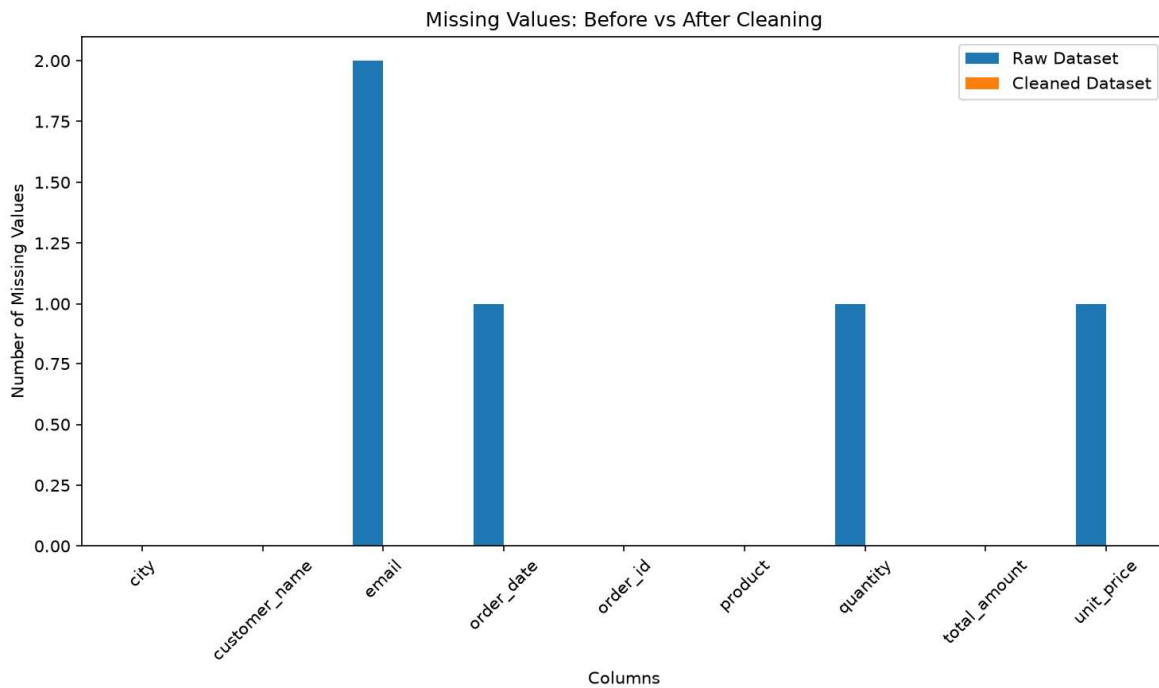


Figure 3: Missing values before and after cleaning

10. Cleaning Execution Output

The cleaning process was executed successfully from the terminal. The following screenshot records the execution and validation output.

```
=====
CLEANING SUMMARY
=====
Original rows: 30
Final rows: 29
Rows removed: 1
Original columns: 8
Final columns: 9
Missing values remaining: 0
Duplicate rows remaining: 0
=====
DATA CLEANING COMPLETED SUCCESSFULLY
=====
PS C:\Users\satyam kumar\Desktop\Data-Cleaning-and-Preparation> |
```

Figure 4: Data cleaning execution output

11. Validation

- Final missing value count: 0.
- Final exact duplicate count: 0.
- Final dataset shape: 29 rows × 9 columns.
- Quantity values are stored as integers.
- Unit prices are stored as floating-point numbers.
- The total_amount field is calculated from quantity × unit_price.
- Dates are standardized to YYYY-MM-DD format.
- Invalid email records are handled safely.

12. Tools and Libraries

Tool / Library	Purpose
Python	Programming language
Pandas	Data loading, cleaning and analysis
NumPy	Numerical operations
Matplotlib	Data visualization
python-docx	Report generation
Git	Version control
GitHub	Source code hosting
VS Code / Terminal	Development environment

13. Project Structure

Data-Cleaning-and-Preparation/

```
|
|— data/
|   |— raw_dataset.csv
|   |— cleaned_dataset.csv
|
|— report/
|   |— Data_Cleaning_Report.docx
|
|— screenshots/
|   |— cleaning_output.png
|   |— missing_values_comparison.png
|   |— product_sales.png
|   |— city_sales.png
|
|— src/
|   |— data_cleaning.py
|   |— data_analysis.py
|
|— analysis_summary.txt
|— cleaning_log.txt
|— README.md
|— requirements.txt
|— .gitignore
```

14. Setup and Usage

Install the required Python libraries using:

```
pip install -r requirements.txt
```

Run the data cleaning process:

```
python src/data_cleaning.py
```

Run the analysis and visualization process:

```
python src/data_analysis.py
```

15. External Libraries, APIs, Datasets and Resources

- Dataset: Synthetic customer sales dataset created specifically for this project.
- Pandas: Used for data manipulation and cleaning.
- NumPy: Used for numerical data processing.
- Matplotlib: Used for charts and visualizations.
- python-docx: Used to generate this project report.
- No external API is required for this project.
- No private credentials, passwords or API keys are included.

16. Security and Privacy

The dataset used in this project is synthetic and does not represent real customer records. The repository does not contain passwords, API keys, private credentials or other sensitive authentication data.

17. Conclusion

This project demonstrates a complete data cleaning and preparation workflow using Python. The raw dataset contained realistic data-quality issues such as missing values, invalid emails, inconsistent formatting, multiple date formats, negative numerical values and a business-key duplicate.

After applying systematic cleaning and validation techniques, the dataset was transformed into a consistent 29-row, 9-column dataset with zero remaining missing values and zero exact duplicate rows. The cleaned data was then used to produce product-wise and city-wise sales analysis.

The project provides a reproducible foundation for further analytics and demonstrates practical data preparation skills using Python.

18. GitHub Repository

<https://github.com/rahulraj2007k-pixel/Data-Cleaning-and-Preparation>

19. Author

Student Project — Data Cleaning and Preparation

The project source code, datasets, analysis files, screenshots and documentation are maintained in the GitHub repository listed above.